



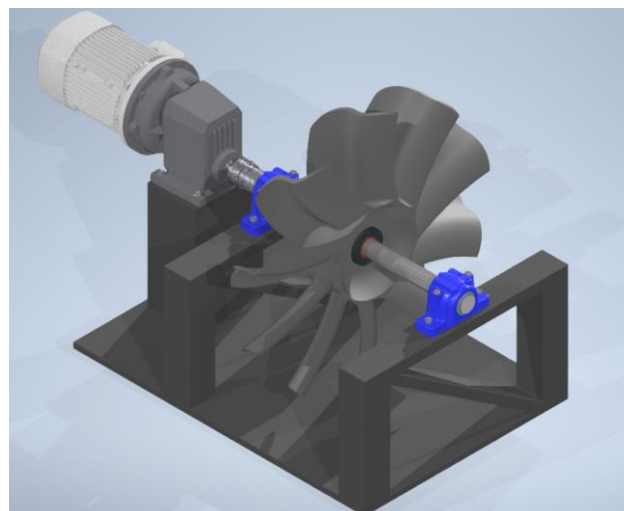
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**Replacing a belt drive system with a gearbox for an
industrial extractor fan**



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Basic data:

Type of fan: V1 (Internal impeller)

Power $P_{jk} = P_{\text{working}} = 3\text{kW}$

Operating speed of the fan $n_{jk} = 1400\text{RPM}$

Operating speed of the motor $n_{\text{mot}} = \text{RPM}$

Thrust (axial) load on the rotor = $F_{ax} = 470\text{N}$

Radial load on the rotor = $F_R = 985\text{N}$

Mass of the rotor $m_{jk} = 44\text{kg}$

Torque of the fan $M = 26.6\text{Nm}$

Geometrical data:

Distance of the bearings from the rotor $a = 300\text{mm}$

Shaft diameter under the rotor $d = 48\text{mm}$

Selecting a new motor:

The motor selected will be the BE 112M model by Bonfiglioli. It has 2 poles, and a synchronous speed of $n_s = 2920\text{RPM}$. This gives it an asynchronous speed of approximately 2800 RPM.

Assuming a gearbox efficiency of 96% (which is typical for helical or bevel geared reducers), the minimum required output power of the motor is given by:

$$P_{\text{power}} = \frac{P_{\text{working}}}{\eta} = \frac{3000}{0.96} = 3125\text{W} = 3.125\text{kW}$$

The nominal power P_n of 4kW is greater than P_{power} , therefore the motor is suitable for the application.

The table below shows the specifications of the newly chosen motor.

2 P		3000 rpm - S1													50 Hz - IE2		
P _n			n	T _n	I _n 400V	η%			cos φ	I _s I _n	T _s T _n	T _a T _n	KVA Code	J _m	IM B5		
HP	kW					rpm	lb-in	A							100%	75%	50%
1	0.75	BE 80A	2	2860	22.1	1.65	80	79.6	76.4	0.83	6.8	3.8	3.5	J	0.0214	9.5	21
1.5	1.1	BE 80B	2	2845	32.7	2.35	81.5	82.2	79.9	0.83	6.9	3.8	3.1	J	0.0271	11.3	25
2	1.5	BE 90SA	2	2865	44.3	3.2	81.3	80.7	78.1	0.82	6.8	3.6	2.8	J	0.0297	12.3	27
3	2.2	BE 90L	2	2870	65	4.7	83.2	83.1	80.8	0.82	6.9	3.1	2.9	J	0.0396	14	31
4	3	BE 100L	2	2880	88	6.2	84.6	84.6	83.7	0.83	7.3	3.5	3.1	J	0.0925	23	51
5.5	4	BE 112M	2	2920	116	8.2	85.8	85.5	84.3	0.82	7.9	3.5	3.1	K	0.1353	28	62
7.5	5.5	BE 132SA	2	2925	159	10.6	87	85	81.7	0.86	8.5	3.6	3.3	K	0.3441	42	93
10	7.5	BE 132SB	2	2935	212	14.3	88.1	87.4	84.7	0.86	8.8	3.9	3.6	K	0.4224	53	117
12.5	9.2	BE 132MB	2	2920	266	16.4	88.8	86.5	84.2	0.91	8.4	3.7	3.3	J	0.4983	65	143
15	11	BE 160MA	2	2940	319	20	89.4	89.5	88	0.89	8.1	3	2.9	J	0.8068	84	185
20	15	BE 160MB	2	2950	434	27.2	90.5	90.5	89.5	0.88	8.5	3	2.8	K	0.9967	97	214
25	18.5	BE 160L	2	2945	531	32	90.9	90.5	89.8	0.91	7.7	2.9	2.7	H	1.1628	109	240



Estimating the Rotational moment of inertia and finding the service factor:

$$J_{\text{hub}} = \frac{1}{3}mr^2$$

$$J_{\text{impeller}} = \sum_{k=1}^n J_{\text{blade}}$$

Where n is the number of blades, and the kth blade ranges up from 1 to n.

$$J_{\text{working machine}} = J_{\text{impeller}} + J_{\text{hub}} = 324542.852 \text{ kgmm}^2 = 0.324543 \text{ kgm}^2$$

(obtained from the Autodesk Inventor model)

The operating speed of the motor is 2800RPM and the required speed of the fan is 1400RPM. This gives a gear ratio of 2:1, therefore i = 2. The reflected inertia, which is the effective inertia experienced by the motor is given by:

$$J_1 = \frac{J_{\text{working machine}}}{i^2} = 0.08114 \text{ kgm}^2$$

The rotor's moment of inertia J_m is 0.1353 lb·ft². This is equivalent to 0.057016 kgm². The acceleration factor is given by:

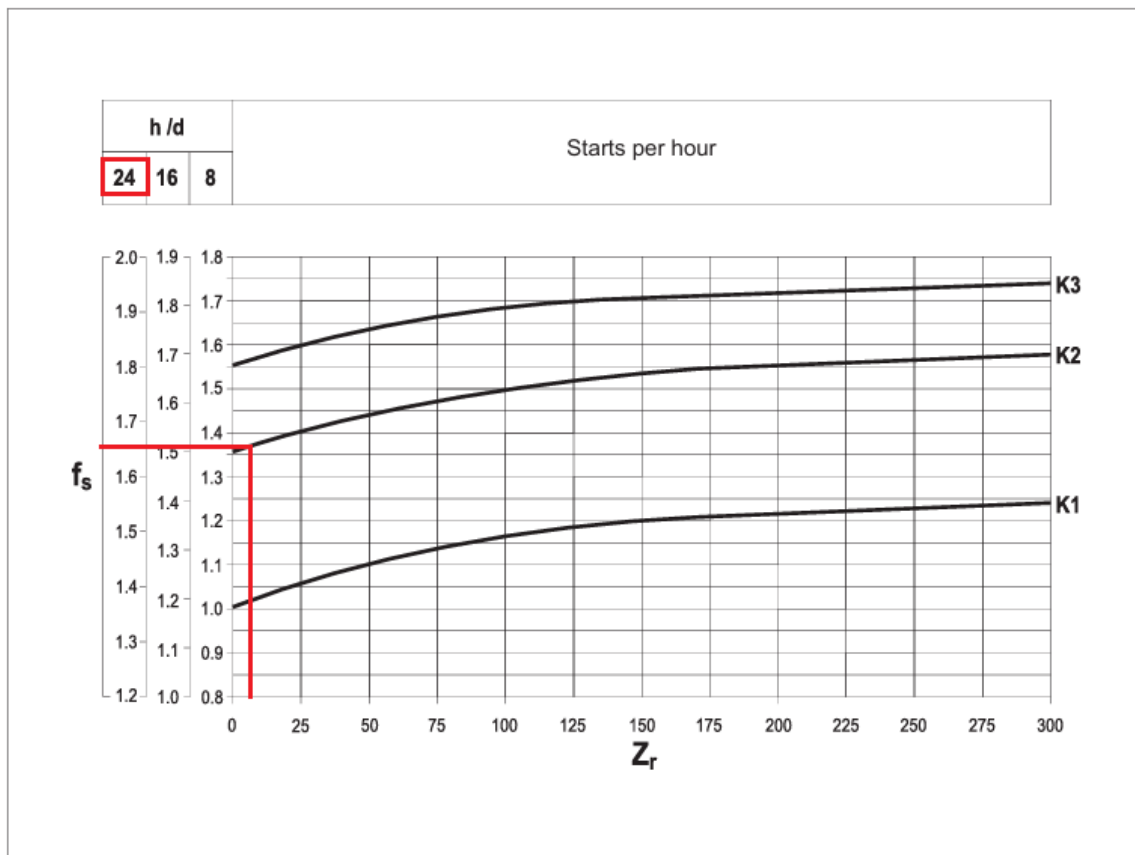
$$K = \frac{J_1}{J_m} = 1.423$$

This places the load into the category of moderate shock, according to the acceleration factor table below.

$K \leq 0,25$	→	K1	Carico uniforme	Uniform load	Gleichmäßige Belastung	Charge uniforme
$0,25 < K \leq 3$	→→	K2	Carico con urti moderati	Moderate shock load	Belastung mit mäßigen Stößen	Charge avec chocs modérés
$3 < K \leq 10$	→	K3	Carico con forti urti	Heavy shock load	Belastung mit starken Stößen	Charge avec chocs importants

This allows for the determination of the service factor, which is as shown below.

The motor operates for 18 or more hours per day, which on the table from the Bonfiglioli catalogue shown below, puts into the $h/d = 24$ category. Z_r is the number of starts per hour. Since the motor is used for an extractor fan that is likely to be running continuously, often several hours without stopping. Therefore, Z_r will be assumed to be no more than 5.



Therefore, the service factor $f_s = 1.65$.

Selecting a gearbox:

The gearbox selected is the S 40 1 1.9 model by Bonfiglioli. The gear ratio i is 1.9, which is sufficiently close to 2, the ratio required for the motor to drive the fan. The minimum power and torque the gearbox should be able to operate at is given by:

$$M_{e2} = f_s M_g = 43.89 \text{ Nm}$$

$$P_{e1} = f_s P_{\text{power}} = 5.16 \text{ kW}$$

The nominal torque M_{n2} and power P_{n1} supported by the gearbox are 48Nm and 7.8kW, respectively. They are both greater than their respective required values, thus the gearbox is suitable. The table below shows the specifications of the gearbox.

S 40		125 Nm										
	i	n ₁ = 2800 min ⁻¹					n ₁ = 1400 min ⁻¹					
		n ₂ min ⁻¹	M _{n2} Nm	P _{n1} kW	R _{n1} N	R _{n2} N	n ₂ min ⁻¹	M _{n2} Nm	P _{n1} kW	R _{n1} N	R _{n2} N	
S 40 1_1.4	1.4	2059	48	10.6	2000	1270	1029	60	6.6	2000	1600	509
S 40 1_1.9	1.9	1514	48	7.8	2000	1450	757	60	4.9	2000	1830	
S 40 1_2.4	2.4	1172	70	8.8	1860	1490	586	90	5.6	2000	1870	
S 40 1_3.1	3.1	918	70	6.9	2000	1660	459	90	4.4	2000	2080	
S 40 1_3.8	3.8	735	70	5.5	2000	1830	367	90	3.5	2000	2290	
S 40 1_4.8	4.8	580	70	4.3	2000	2020	290	90	2.8	2000	2530	
S 40 1_6.1	6.1	461	70	3.5	2000	2220	231	90	2.2	2000	2790	
S 40 1_7.2	7.2	392	63	2.6	2000	2410	196	80	1.7	2000	3030	
S 40 1_8.6	8.6	324	48	1.7	2000	2670	162	60	1.0	2000	3370	
S 40 1_10.7	10.7	262	40	1.1	2000	2930	131	50	0.70	2000	3690	
S 40 1_12.4	12.4	226	40	1.0	2000	3100	113	50	0.60	2000	3800	

Selecting a coupling:

The output shaft's diameter of the gearbox is 28mm, while the fan drive shaft will have a 40mm diameter at its connecting end. A ROTEX jaw coupling with a ROTEX size of 28 (the size designation system ROTEX uses to match compatible hubs) will be used. The size designation of the spider between the hubs will also have a ROTEX size of 28. Since the fan is not expected to experience significant vibrations, the rated torque of 95Nm for the spider is sufficient to gauge its suitability. Since this value is over twice the required torque value of $M_{e2} = 43.89$, it can easily handle the expected load on it.

The two hubs of the coupling will have 28mm and 40mm bores to match their respective shafts. Torque will be transmitted through feather key and groove connections, using DIN 6885 H7 keyways. The two tables below show the specifications of the hubs and spider.

ROTEX® Flexible jaw couplings

Cylindrical bores and spline bores

Stock programme of cylindrical finish bores [mm] H7 feather keyway to DIN 6885 sheet 1 [JS9] and thread for setscrews		Un- bo- red	Ø6	Ø8	Ø9	Ø10	Ø11	Ø12	Ø14	Ø15	Ø16	Ø17	Ø18	Ø19	Ø20	Ø22	Ø24	Ø25	Ø28	Ø30	Ø32	Ø35	Ø38	Ø40	Ø42	Ø45	Ø48	Ø50	Ø55	Ø60	Ø65	Ø70	Ø75	Ø80	Ø85	Ø90	Ø100
ROTEX® Size/material																																					
14	Sint	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•
	Al-H	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•
19	Sint	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•
	AL-D	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•
24	Sint	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•
	Al-D	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•
28	Sint	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•
	Al-D	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•
38	Sint	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•
	GJL	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•
42	Sint	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•
	GJL	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•
48	Sint	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•
	GJL	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•
55	Sint	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•
	GJL	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•
65	Sint	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•
	GJL	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•
75	Sint	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•
	GJL	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•
90	Sint	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•
	GJL	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•
	St	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•

Technical data of standard spiders

92 Shore-A spider made of T-PUR® and PUR															
ROTEX® size	Max. speed		Twist angle φ with		Torque [Nm]			Damping power PKW [W/1)	Relative damping ψ	Resonance factor VR	Torsion spring stiffness C dyn. [Nm/rad]				
	V=35 m/s casted material	V=40 m/s steel	TKN	TK max	Rated (TKN)	Max (TK max)	Vibratory (TKW)				1.0 TKN	0.75 TKN	0.5 TKN	0.25 TKN	
14	22200	25400	6,4°	10°	7,5	15	2,0	–			0,38x10³	0,31x10³	0,24x10³	0,14x10³	
19	16700	19000			10	20	2,6	4,8			1,28x10³	1,05x10³	0,80x10³	0,47x10³	
24	12100	13800			35	70	9,1	6,6			4,86x10³	3,98x10³	3,01x10³	1,79x10³	
28	10100	11500			95	190	25	8,4			10,90x10³	8,94x10³	6,76x10³	4,01x10³	
38	8300	9500			190	380	49	10,2			21,05x10³	17,26x10³	13,05x10³	7,74x10³	
42	7000	8000			265	530	69	12,0			23,74x10³	19,47x10³	14,72x10³	8,73x10³	
48	6350	7250			310	620	81	13,8			36,70x10³	30,09x10³	22,75x10³	13,49x10³	
55	5550	6350	3,2°	5°	410	820	107	15,6		0,80	7,90	50,72x10³	41,59x10³	31,45x10³	18,64x10³
65	4950	5650			625	1250	163	18,0	97,13x10³			79,65x10³	60,22x10³	35,70x10³	
75	4150	4750			1280	2560	333	21,6	113,32x10³			92,92x10³	70,26x10³	41,65x10³	
90	3300	3800			2400	4800	624	30,0	190,09x10³			155,87x10³	117,86x10³	69,86x10³	
100	2950	3350			3300	6600	858	36,0	253,08x10³			207,53x10³	156,91x10³	93,01x10³	
110	2600	2950			4800	9600	1248	42,0	311,61x10³			255,52x10³	193,20x10³	114,52x10³	
125	2300	2600			6650	13300	1729	48,0			474,86x10³	389,39x10³	294,41x10³	174,51x10³	
140	2050	2350			8550	17100	2223	54,6			660,49x10³	541,60x10³	409,50x10³	242,73x10³	
160	1800	2050			12800	25600	3328	75,0			890,36x10³	730,10x10³	552,03x10³	327,21x10³	
180	1550	1800			18650	37300	4849	78,0			2568,56x10³	2106,22x10³	1592,51x10³	943,95x10³	

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